

- Built backend APIs using **Spring Boot (Java)** with role-based access control for Admin, Librarian, and Student roles.
 - Implemented a **Node.js + Express** backend alternative with RESTful services and JWT-based authentication.
 - Developed a responsive frontend using **React.js**, enabling real-time book availability, issue tracking, and user dashboards.
 - Integrated relational databases (MySQL/PostgreSQL) for transactional consistency and optimized queries.
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Food Spoilage Detection System : IoT Project | Raspberry Pi & Arduino Nano 33 IoT

- Built an **IoT-based food management system** to predict food spoilage using environmental sensor data.
 - Collected temperature, humidity, and gas sensor readings via **Arduino Nano 33 IoT**, processed using **Raspberry Pi**.
 - Implemented logic to analyze spoilage thresholds and notify users before food becomes unsafe.
 - Demonstrated real-world application of embedded systems, sensor integration, and data-driven decision making.
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Smart Gate Pass Management System: IoT + Full-Stack Application

- Developed a **hostel gate pass system** integrating IoT hardware with a web-based frontend.
 - Enabled students to apply for gate passes online; wardens approve/reject requests via an admin dashboard.
 - Integrated **RFID-based access control** at the gate to allow exit only after approval.
 - Used **Raspberry Pi and Arduino Nano 33 IoT** for hardware integration and secure validation.
 - Improved security, reduced manual effort, and digitized the complete leave approval workflow.
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Real-Time Task Management System: Backend-Focused Project | Node.js, WebSockets, MongoDB

- Built a real-time **task assignment and collaboration system** using **Node.js** and **WebSockets**.
 - Enabled users to assign tasks, update statuses, and receive live updates without page refresh.
 - Designed MongoDB schemas and implemented **database sharding** for scalability.
 - Developed backend APIs with authentication, authorization, and data validation.
 - Used basic HTML/CSS frontend to demonstrate backend-first, system-level design.
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Employability Skills & Leadership

- **Teamwork:** Collaborated effectively on group projects and academic coursework.
- **Problem Solving:** Applied analytical thinking to solve complex coding and algorithmic challenges.
- **Communication:** Strong written and verbal communication through presentations and technical documentation.
- **Engineering Discipline:** Clean code practices, testing mindset, structured and maintainable development.